Differentiability

Week 9, Chapter 5

Definition of Derivatives

Definition 2.1 Differentiability

Let $f : \left(a, b\right) \to \mathbb{R}$ be a function. Then we say that f is differentiable at a point $c \in \left(a, b\right)$ if

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c} \text{ exists.}$$

$f'(c)$ is called as the derivative of f at c.

Remark: Alternative definition

Let $x = c + h$. Then f is differentiable at c if

$$f'(c) = \lim_{h \to 0} \frac{f(c+h) - f(c)}{h} \text{ exists.}$$

This formula is normally used to find the (first) derivative of a function (the previous one is used to determine the differentiability).

Remark: Sequential definition

Let $f : \left(a, b\right) \to \mathbb{R}$ and $c \in \left(a, b\right)$. Then we can construct a sequence $\left(x_n\right)$ which converges to c, and define f is differentiable at c if

$$f'(c) = \lim \left(\frac{f(x_n) - f(c)}{x_n - c} \right) \text{ exists.}$$

The definitions of derivative of *f* at *c*

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c}$$

$$f'(c) = \lim_{h \to 0} \frac{f(c+h) - f(c)}{h}$$

$$f'(c) = \lim_{n \to \infty} \frac{f(x_n) - f(c)}{x_n - c}$$

Derivative at endpoints

Let $f : \left[a, b\right] \to \mathbb{R}$ be a function. Then we say that f is differentiable at the endpoints a and b if $f'(a) = \lim\limits_{x \to a^+} \frac{f(x) - f(a)}{x - a}$ and $f'(b) = \lim\limits_{x \to b^-} \frac{f(x) - f(b)}{x - b}$ exist.

Derivatives on an interval

Let $f : D \to \mathbb{R}$ be a function. Then we say that f is differentiable on D if f is differentiable at all points in D.

Example 2.3

Let $f, g, k : \mathbb{R} \to \mathbb{R}$ be functions defined by

$$f(x) = 1, \;\; g(x) = x, \;\; k(x) = x^2 \, .$$

Let $c \in \mathbb{R}$. Then

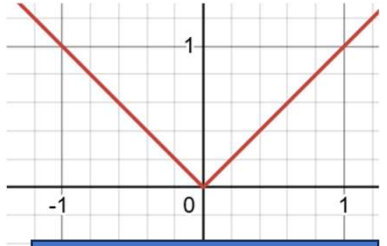
$$f'(c) = \lim_{h \to 0} \frac{f(c+h) - f(c)}{h} = \lim_{h \to 0} \frac{1-1}{h} = 0$$

$$g'(c) = \lim_{h \to 0} \frac{g(c+h) - g(c)}{h} = \lim_{h \to 0} \frac{c+h-c}{h} = 1$$

$$k'(c) = \lim_{h \to 0} \frac{(c+h)^2 - c^2}{h} = \lim_{h \to 0} \frac{2ch - h^2}{h} = 2c$$

Example 3.2

Let $f : \mathbb{R} \to \mathbb{R}$ be defined as $f(x) = |x|$.



'sharp' point at 0

Show that f is not differentiable at 0.

Proof. $\lim\limits_{x\to 0^-} \frac{f(x)-f(0)}{x-0} = \lim\limits_{x\to 0^-} \frac{-x}{x} = -1 \neq 1 = \lim\limits_{x\to 0^+} \frac{x}{x} = \lim\limits_{x\to 0^+} \frac{f(x)-f(0)}{x-0}$,

so $\lim\limits_{x\to 0} \frac{f(x)-f(0)}{x-0}$ does not exist. Thus, f is not differentiable at 0.

Extra Example

Let $f : \mathbb{R} \to \mathbb{R}$ be defined as $f(x) = \begin{cases} x^3, x < 0 \\ x^2, x \geq 0 \end{cases}$.

Is f differentiable (and continuous) at 0?

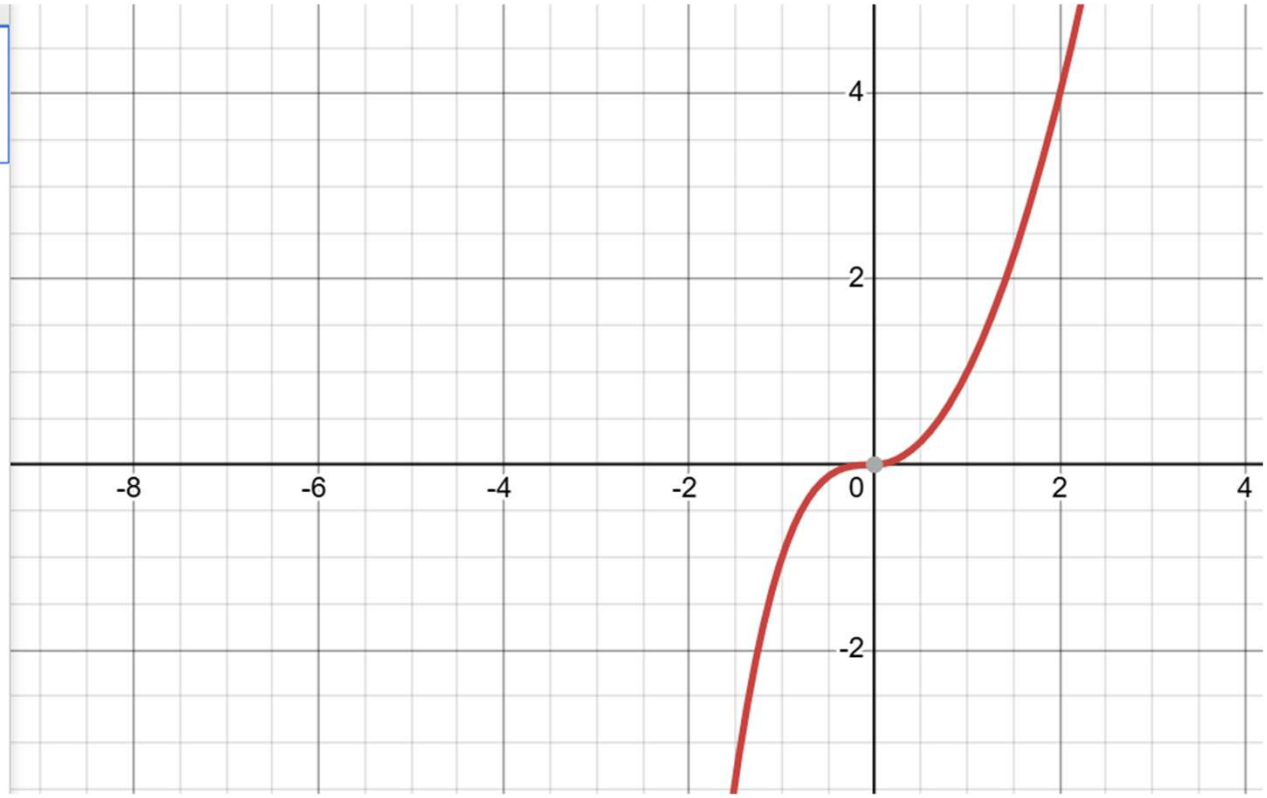
Proof. $\lim\limits_{x \to 0^-} \frac{f(x)-f(0)}{x-0} = \lim\limits_{x \to 0^-} \frac{x^3}{x} = 0 = \lim\limits_{x \to 0^+} \frac{x^2}{x} = \lim\limits_{x \to 0^+} \frac{f(x)-f(0)}{x-0}$.

Thus, the limit exists and $f'(0) = 0$.

So, f is differentiable (and continuous) at 0.

$$y = \left[x^3\{x < 0\}, x^2\{x \geq 0\} \right]$$

My son showed me how to type piecewise function.



Extra Example

Let $f : \mathbb{R} \to \mathbb{R}$ be defined as $f(x) = \begin{cases} (x-4)^2, x < 4 \\ 0, \quad x \geq 0 \end{cases}$.

Show that f is differentiable at 4.

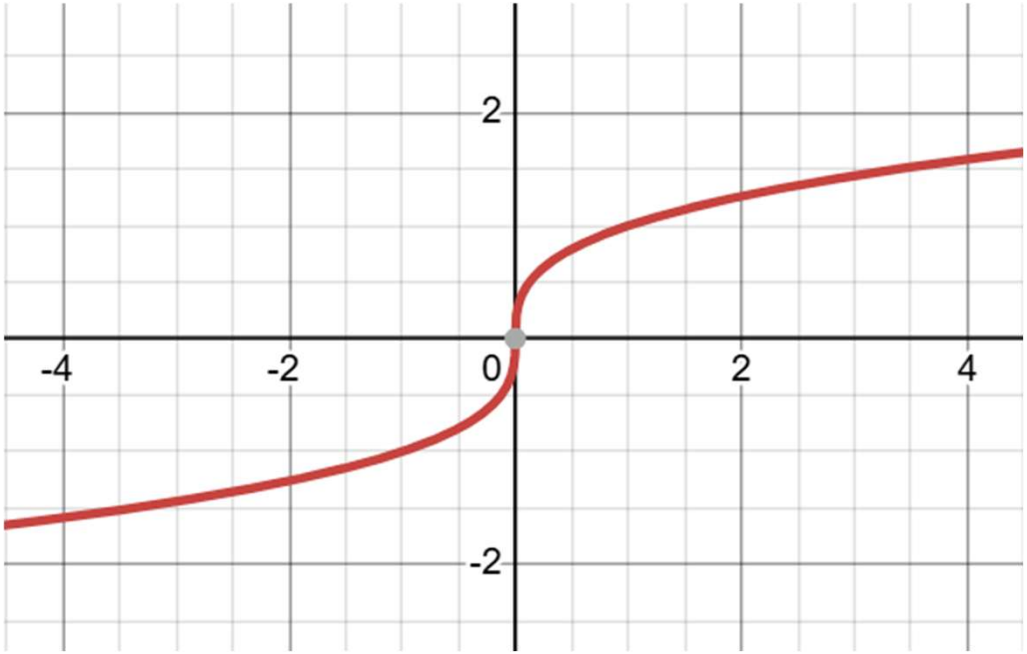
Proof. $\lim\limits_{x \to 4^-} \frac{(x-4)^2 - 0}{x-4} = \lim\limits_{x \to 4^-} (x-4) = 0 = \lim\limits_{x \to 4^+} \frac{0}{x-4}$.

So, f is differentiable at 4.

Exercise

Let $f : \mathbb{R} \to \mathbb{R}$ be defined as $f(x) = x^{\frac{1}{3}}$.

Is f differentiable at 0?



True or False

All three definitions of derivative are taking limit

$$\lim_{x\to\square} \qquad \lim_{h\to\square} \qquad \lim_{n\to\square}$$

to the same value. False.

True or False

Geometrically, if a sharp edge can be seen at c, then f is not differentiable at c.

But the converse is false.

Counter example: $f(x) = x^{\frac{1}{3}}$ at 0 is not sharp but it is not differentiable at 0.

Properties of Derivatives

Theorem 3.1

Let $f : \big(a, b\big) \to \mathbb{R}$ be a function.

If f is differentiable at $c \in \big(a, b\big)$, then f is continuous at c.

Proof. $$\begin{aligned} \lim_{x \to c} \Big(f(x) - f(c) \Big) &= \lim_{x \to c} \frac{\big(f(x) - f(c)\big)(x - c)}{x - c} \\ &= \lim_{x \to c} \frac{f(x) - f(c)}{x - c} \times \lim_{x \to c} (x - c) \\ &= f'(c) \times 0 = 0 \end{aligned}$$

Theorem 3.3, Theorem 3.4, Theorem 3.6

Let $\lambda \in \mathbb{R}$. Let $f, g : \Big(a, b\Big) \to \mathbb{R}$ be differentiable functions at $c \in \Big(a, b\Big)$.

Then λf, $f + g$, fg, $\frac{f}{g}$ are differentiable functions at c, with derivatives

$$\Big(\lambda f\Big)' = \lambda f' \text{ (scalar product rule)}, \Big(f + g\Big)' = f' + g' \text{ (sum rule)},$$

$$\Big(fg\Big)' = f'g + fg' \text{ (product rule), and } \Big(\frac{f}{g}\Big)' = \frac{f'g - g'f}{g^2}, g \neq 0 \text{ (quotient rule).}$$

Example 3.5 (Derivative of polynomial)

Any polynomial $p(x) = \sum_{r=0}^{n} a_r x^r$, $a_n \neq 0$ is differentiable on $\mathbb{R}$, with $p'(x) = \sum_{r=1}^{n} r a_r x^{r-1}$.

Proof. Exercise.

Theorem 3.7 (The chain rule)

Let $f : D_1 \to \mathbb{R}$ and $g : D_2 \to \mathbb{R}$ with $f(D_1) \subseteq D_2$.

Suppose f is differentiable at $c \in D_1$ and

g is differentiable at $f(c) \in f(D_1)$. Then

$g \circ f : D_1 \to \mathbb{R}$ is differentiable at c with

derivative $\left(g \circ f\right)'(c) = g'(f(c)) \cdot f'(c)$.

True or False

If f is continuous at c, then f is differentiable at c.

False. Counter example: $f(x) = x^{\frac{1}{3}}$ is continuous at 0 but it is not differentiable at 0.

Proof of Theorems

The scalar product rule

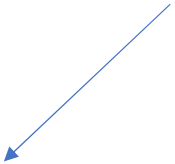
$$\left(\lambda f\right)'(c) = \lim_{x \to c} \frac{\lambda f(x) - \lambda f(c)}{x - c} = \lambda \lim_{x \to c} \frac{f(x) - f(c)}{x - c} = \lambda f'(c)$$

The sum rule

$$\begin{aligned}\left(f+g\right)'(c) &= \lim_{x\to c}\frac{f(x)+g(x)-f(c)-g(c)}{x-c}\\ &= \lim_{x\to c}\frac{f(x)-f(c)}{x-c}+\lim_{x\to c}\frac{g(x)-g(c)}{x-c}\\ &= f'(c)+g'(c)\end{aligned}$$

The product rule

$$
\begin{aligned}
\Big(fg\Big)'(c) &= \lim_{x\to c} \frac{f(x)g(x)-f(c)g(c)}{x-c} \\
&= \lim_{x\to c} \frac{f(x)g(x)-g(x)f(c)+f(c)g(x)-f(c)g(c)}{x-c} \\
&= \lim_{x\to c} \left(g(x)\,\frac{f(x)-f(c)}{x-c} \right) + \lim_{x\to c} \left(f(c)\,\frac{g(x)-g(c)}{x-c} \right) \\
&= g(c)f'(c) + f(c)g'(c)
\end{aligned}
$$



The reciprocal rule

$$\left(\frac{1}{g}\right)'(c) = \lim_{x \to c} \frac{\frac{1}{g(x)} - \frac{1}{g(c)}}{x-c} = \lim_{x \to c} \frac{g(c)-g(x)}{(x-c)g(x)g(c)}$$
$$= -\frac{1}{g(c)} \lim_{x \to c} \frac{g(x)-g(c)}{(x-c)g(x)} = -\frac{g'(c)}{\Big(g(c)\Big)^2}$$

The quotient rule

Apply the product rule and reciprocal rule:

$$\begin{aligned}\left(\frac{f}{g}\right)'(c) &= \left(f \cdot \frac{1}{g}\right)'(c) \\ &= f'(c) \cdot \frac{1}{g(c)} + f(c)\left(\frac{1}{g(c)}\right)' \\ &= \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}\end{aligned}$$

The quotient rule (alternative proof)

$$\begin{aligned}
\left(\frac{f}{g}\right)'(c) &= \lim_{x\to c} \frac{f(x)/g(x) - f(c)/g(c)}{x-c} \\
&= \lim_{x\to c} \frac{f(x)g(c) - f(c)g(c) + f(c)g(c) - f(c)g(x)}{(x-c)g(x)g(c)} \\
&= \lim_{x\to c} \frac{1}{g(x)g(c)} \left(g(c)\frac{f(x)-f(c)}{x-c} - f(c)\frac{g(x)-g(c)}{x-c} \right) \\
&= \frac{f'(c)g(c) - f(c)g'(c)}{\big(g(c)\big)^2}
\end{aligned}$$

The chain rule (problem!!)

$$\begin{aligned}\left(g \circ f\right)'(c) &= \lim_{x \to c} \frac{g(f(x)) - g(f(c))}{x - c} \\ &= \lim_{x \to c} \left(\frac{g(f(x)) - g(f(c))}{f(x) - f(c)} \cdot \frac{f(x) - f(c)}{x - c} \right) \\ &= g'(f(c)) \cdot f'(c), \quad \text{is } f(x) \neq f(c)\,?\end{aligned}$$

Cannot do like this!!

The chain rule

Proof. Let $f : D_1 \to \mathbb{R}$ and $g : D_2 \to \mathbb{R}$, $f(D_1) \subseteq D_2$.

Let $d : D_2 \to \mathbb{R}$ with $d(y) = \begin{cases} \frac{g(y)-g\left(f(c)\right)}{y-f(c)}, y \neq f(c) \\ g'\left(f(c)\right), y = f(c) \end{cases}$.

Then $\forall x \in D_1 \setminus \{c\}$, $\exists y = f(x) \in f(D_1)$ such that

$$d(y) = d\left(f(x)\right) = \frac{g\left(f(x)\right)-g\left(f(c)\right)}{f(x)-f(c)} \text{ and}$$

$$d\left(f(x)\right) \times \frac{f(x)-f(c)}{x-c} = \frac{g\left(f(x)\right)-g\left(f(c)\right)}{f(x)-f(c)} \times \frac{f(x)-f(c)}{x-c} = \frac{g\left(f(x)\right)-g\left(f(c)\right)}{x-c}.$$

$$
\begin{aligned}
\Big(g \circ f\Big)'(c) &= \lim_{x \to c} \frac{g\Big(f(x)\Big) - g\Big(f(c)\Big)}{x - c} \\
&= \lim_{x \to c} d\Big(f(x)\Big) \times \frac{f(x) - f(c)}{x - c} \\
&= \lim_{x \to c} d\Big(f(x)\Big) \times \lim_{x \to c} \frac{f(x) - f(c)}{x - c} \\
&= d\Big(f(c)\Big) \times f'(c) \\
&= g'\Big(f(c)\Big) f'(c)
\end{aligned}
$$

Interior Extremum Theorem

Recap

Theorem 8.1 (Chapter 4). Boundedness Theorem.

Let $f : \left[a, b\right] \to \mathbb{R}$ be continuous on $\left[a, b\right]$.

Then f is bounded.

Theorem 8.2 (Chapter 4). Extreme Value Theorem

Let $f : \left[a, b\right] \to \mathbb{R}$ be continuous on $\left[a, b\right]$.

Then f attains its bounds on $\left[a, b\right]$.

Theorem 4.1. Interior Extremum Theorem (Fermat's Theorem)

Let $f : \big(a,b\big) \to \mathbb{R}$ be differentiable on $\big(a,b\big)$.

If f attains a maximum or minimum at $c \in \big(a,b\big)$, then $f'(c) = 0$.

Remark: The converse of the theorem is false.

Proof. Because $c \in \left(a, b\right)$, we can construct two sequences $\left(x_n\right)$ and $\left(y_n\right)$, which converge to c and satisfy

$$x_n < c < y_n \forall n \in \mathbb{N}.$$

The fact that $f_{\max} = f(c)$ implies that $f(y_n) - f(c) \leq 0 \ \forall n \in \mathbb{N}$,

and thus $f'(c) = \lim_{n\to\infty} \frac{f(y_n)-f(c)}{y_n-c} \leq 0$

by the Order Limit Theorem (Theorem 6.2, Chap 2).

In a similar way, $\frac{f(x_n)-f(c)}{x_n-c} \geq 0\,,\ x_n \in \Big(a,b\Big)\ \forall n \in \mathbb{N}$

because both numerator and denominator are negative.

This implies that $f'(c) = \lim\limits_{n\to\infty} \frac{f(x_n)-f(c)}{x_n-c} \geq 0\,.$

Hence, $f'(c) = 0\,.$

The proof for when $f_{\min} = f(c) \Rightarrow f'(c) = 0$ is similar. (Exercise).

True or False

Let $f:\left(a,b\right)\to\mathbb{R}$ be differentiable on $\left(a,b\right)$.

If $f'(c)=0$, then f attains a maximum or minimum at $c\in\left(a,b\right)$.

False. Counter example: Let $f(x)=x^3$. Then $f'(0)=0$

but f has no maximum nor minimum.

Definition: Local extrema

Let $f : \left[a, b\right] \to \mathbb{R}$ be continuous on $\left[a, b\right]$ and differentiable on $\left(a, b\right)$. Let $c \in \left(a, b\right)$. Then $\left(c, f(c)\right)$ is a local minimum if $f(c) < f(c^{\pm})$, and $\left(c, f(c)\right)$ is a local maximum if $f(c) > f(c^{\pm})$.

Remark: $c^{\pm} := c \pm \delta$ where $\delta > 0$ is a significant small value.

Theorem: Let $f : [a, b] \to \mathbb{R}$ be twice differentiable on (a, b).

(a) If $f'(c) = 0$ and $f''(c) > 0$, then $\big(c, f(c)\big)$ is a local minimum.

(b) If $f'(c) = 0$ and $f''(c) < 0$, then $\big(c, f(c)\big)$ is a local maximum.

Remark: If f' is differentiable on D, then f is called twice differentiable on D.

Proof. Given $f'(c) = 0$.

(a) Since $0 < f''(c) = \lim_{x \to c} \frac{f'(x) - f'(c)}{x - c} = \lim_{x \to c^+} \frac{f'(x)}{x - c} = \lim_{x \to c^-} \frac{f'(x)}{x - c}$,

it follows that $\forall \varepsilon > 0$ $\exists \delta_1, \delta_2 > 0$ such that

$\left| \frac{f'(x)}{x - c} - f''(c) \right| < \varepsilon$ whenever $c < x < c + \delta_1$ and

$\left| \frac{f'(x)}{x - c} - f''(c) \right| < \varepsilon$ whenever $c - \delta_2 < x < c$.

Then $f''(c) - \varepsilon < \frac{f'(x)}{x-c} < f''(c) + \varepsilon$ whenever $c < x < c + \delta_1$.

Choose $\varepsilon > 0$ such that $f''(c) - \varepsilon > 0$. Since $x - c > 0$,

it follows that $f'(x) > 0$ (from $f''(c) - \varepsilon < \frac{f'(x)}{x-c}$) whenever

$c < x < c + \delta_1$, or $f(x)$ is increasing in $x \in \left(c, c + \delta_1\right)$.

Similarly, we can show that $f(x)$ is decreasing in $x \in \left(c - \delta_2, c\right)$.

Hence, $\left(c, f(c)\right)$ is a local minimum.

(b) Similar arguments to obtain a local maximum.

Darboux’s Theorem

Definition 3.1 (Chapter 4): Darboux's function

Let $f : [a,b] \to \mathbb{R}$ be a function. Then f is called *Darboux* if for all $\alpha \in \big(f(a), f(b)\big)$ or $\alpha \in \big(f(b), f(a)\big)$, there exists $c \in (a,b)$ such that $f(c) = \alpha$.

Remark: Geometrically, Darboux is a function with no gaps in between $f(a)$ and $f(b)$.

Darboux's Theorem.

Let $f : \left[a, b\right] \to \mathbb{R}$ be differentiable on $\left[a, b\right]$.

Let $\alpha \in \Big(f'(a), f'(b)\Big)$ or $\alpha \in \Big(f'(b), f'(a)\Big)$.

Then there exists $c \in \Big(a, b\Big)$ such that $f'(c) = \alpha$.

Proof. Let $g : \left[a, b\right] \to \mathbb{R}$ be defined as $g(x) = f(x) - \alpha x$.

Note that g is differentiable on $\left[a, b\right]$ with

$$g'(x) = f'(x) - \alpha \, .$$

In terms of g, our hypothesis states that $g'(a) < 0 < g'(b)$,

and we want to show that $g'(c) = 0$ for some $c \in \Big(a, b\Big)$.

Since $g'(a) = \lim_{x \to a} \frac{g(x)-g(a)}{x-a} < 0$, it follows that

$$g'(a) - \varepsilon < \frac{g(x)-g(a)}{x-a} < g'(a) + \varepsilon$$

whenever $0 < \left|x - a\right| < \delta$.

Now $g(x) < g(a) + \Big(g'(a) + \varepsilon\Big)(x - a)$. Choose $\varepsilon > 0$ such that $g'(a) + \varepsilon < 0$. Since $x - a > 0$, we have $g(x) < g(a)$ as desired.

A similar argument concludes that $g(y) < g(b)$.

By Theorem 8.2 (Chapter 4), g obtains a minimum on $\Big[a,b\Big]$, but we just showed that a and b are not minima, thus $c \in \Big(a,b\Big)$ for which $g'(c) = 0$ by Theorem 4.1.

Remark: 1. We see that the derivative of a function, even if it is not continuous, has the intermediate value property.

2. $f'(a)f'(b) < 0\,(\text{when } \alpha = 0\,) \Rightarrow \exists c \in \Big(a, b\Big) \ni f'(c) = 0\,.$

3. If a derivative function attains two distinct values $f'(a)$ and $f'(b)$, then it must also attain every value in between.

True or False

If f is differentiable at c, then f' is continuous at c.

False. Counter example: $f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), x \neq 0 \\ 0 \quad\quad\quad , x = 0 \end{cases}$.

Then f is differentiable at 0 but f' is not continuous at 0.

True or False

If f is differentiable at c, then f' is differentiable at c.

False. Counter example: $f(x) = x|x|$.

Then f is differentiable at 0 but f' is not differentiable at 0.

True or False

Let $f : \left[a, b\right] \to \mathbb{R}$ be a real function such that $\forall \alpha \in \mathbb{R}$ in between $f(a)$ and $f(b)$, $\exists c \in \left(a, b\right) \ni f(c) = \alpha$.

Then f is continuous on $\left[a, b\right]$.

False. Counter example: Let $f:\left[-\frac{2}{\pi},\frac{2}{\pi}\right]\to\mathbb{R}$ defined by

$$f(x)=\begin{cases}\sin\left(\frac{1}{x}\right), x\neq 0\\ 0, \qquad\quad x=0\end{cases}.$$

Then for every $\alpha\in(-1,1)$ in between $f\left(-\frac{2}{\pi}\right)$ and $f\left(\frac{2}{\pi}\right)$, there exists a $c\in\left(-\frac{2}{\pi},\frac{2}{\pi}\right)$ with $f(c)=\alpha$. However, f is not continuous at $x=0$ because $\lim\limits_{x\to 0}\sin(\frac{1}{x})\not\exists$.

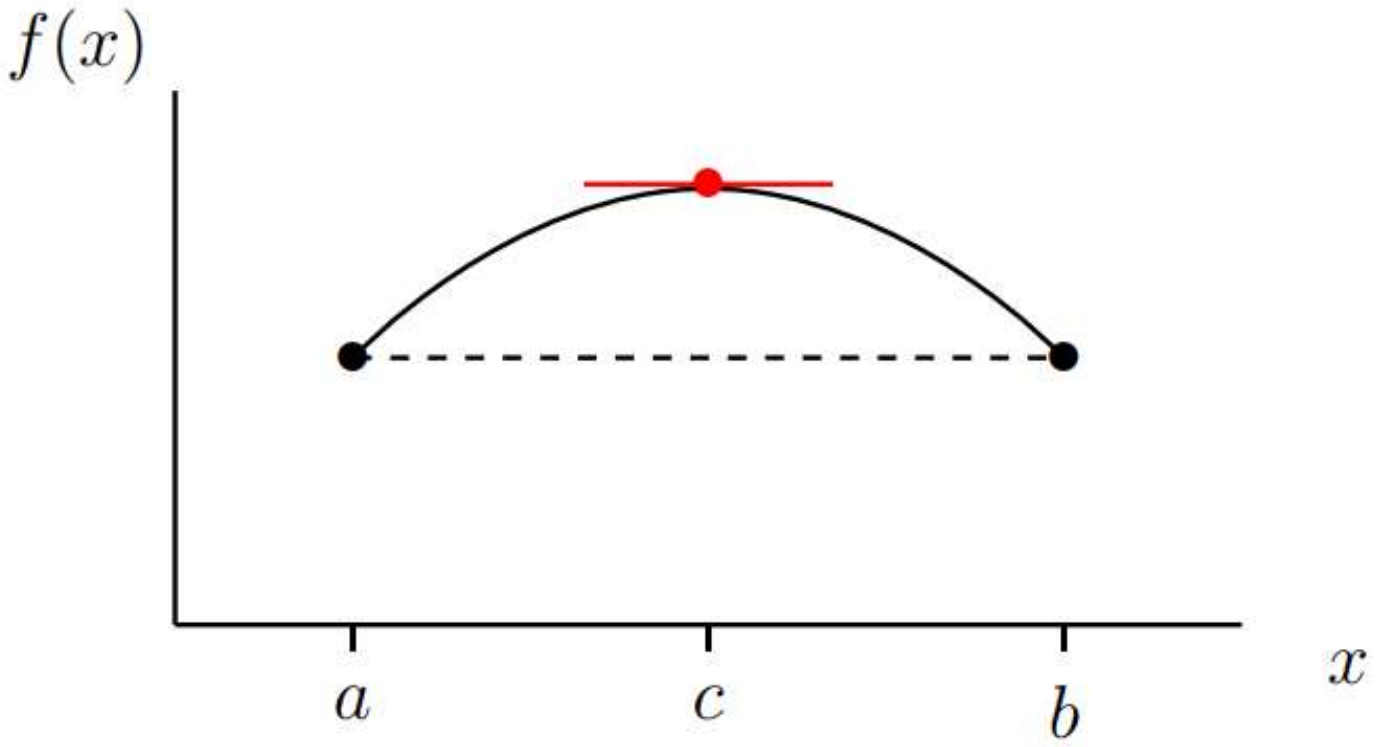
Note that f here is Darboux.

Rolle’s Theorem

Theorem 4.2. Rolle's Theorem

Let $f : \left[a, b\right] \to \mathbb{R}$ be continuous on $\left[a, b\right]$ and differentiable on $\left(a, b\right)$.

Suppose $f(a) = f(b)$, then $\exists c \in \left(a, b\right) \ni f'(c) = 0$.


$f(x)$
a
c
b
x

Proof. By Theorem 8.2, f attains its maximum and minimum on $\left[a,b\right]$.

If the maximum or minimum occurred at $c \in \left(a,b\right)$,

then by Theorem 4.1, $f'(c) = 0$.

Otherwise, f attains its maximum and minimum at both end points.

In this case, f is the constant function on $\left[a,b\right]$, since $f(a) = f(b)$.

Hence, its derivative is zero everywhere on $\left(a,b\right)$.

Applications of Rolle's Theorem

Example

Use the Intermediate Value Theorem and the Rolle's Theorem to prove that $x^5 + x^3 + x = 1$ has exactly one solution in between 0 and 1.

Proof. Let $f(x) = x^5 + x^3 + x - 1$. Then f is continuous on $\left[0,1\right]$ and differentiable on $\left(0,1\right)$. Since $f(0)f(1) < 0$; it follows that there is at least a root, $c \in \left(0,1\right)$ by IVT.

Suppose equation $f(x)=0$ has two roots

$\alpha,\beta\in\left(0,1\right),\ \alpha<\beta$. Then $f(\alpha)=f(\beta)=0$.

By RT, $\exists c\in\left(\alpha,\beta\right)\subset\left(0,1\right)\ni f'(c)=0$; but

$0=f'(c)=5c^4+3c^2+1>0\ \forall c\in\mathbb{R}$, a contradiction.

Example 4.3

Prove that $x^3 + x + 1 = 0$ has exactly one root on $\mathbb{R}$.

Proof. Let $f(x) := x^3 + x + 1 = 0$. Then f is continuous everywhere and differentiable on $\mathbb{R}$. By IVT,

$$f(-1)f(0) = -1 < 0 \Rightarrow \exists c \in \Big(-1, 0\Big) \ni f(c) = 0\,.$$

Now suppose $f(x) = 0$ has two zeros $\alpha, \beta \in \mathbb{R}$.

Then $f(\alpha) = f(\beta) = 0$. By RT, $\exists c \in \mathbb{R}$ э $f'(c) = 0$,

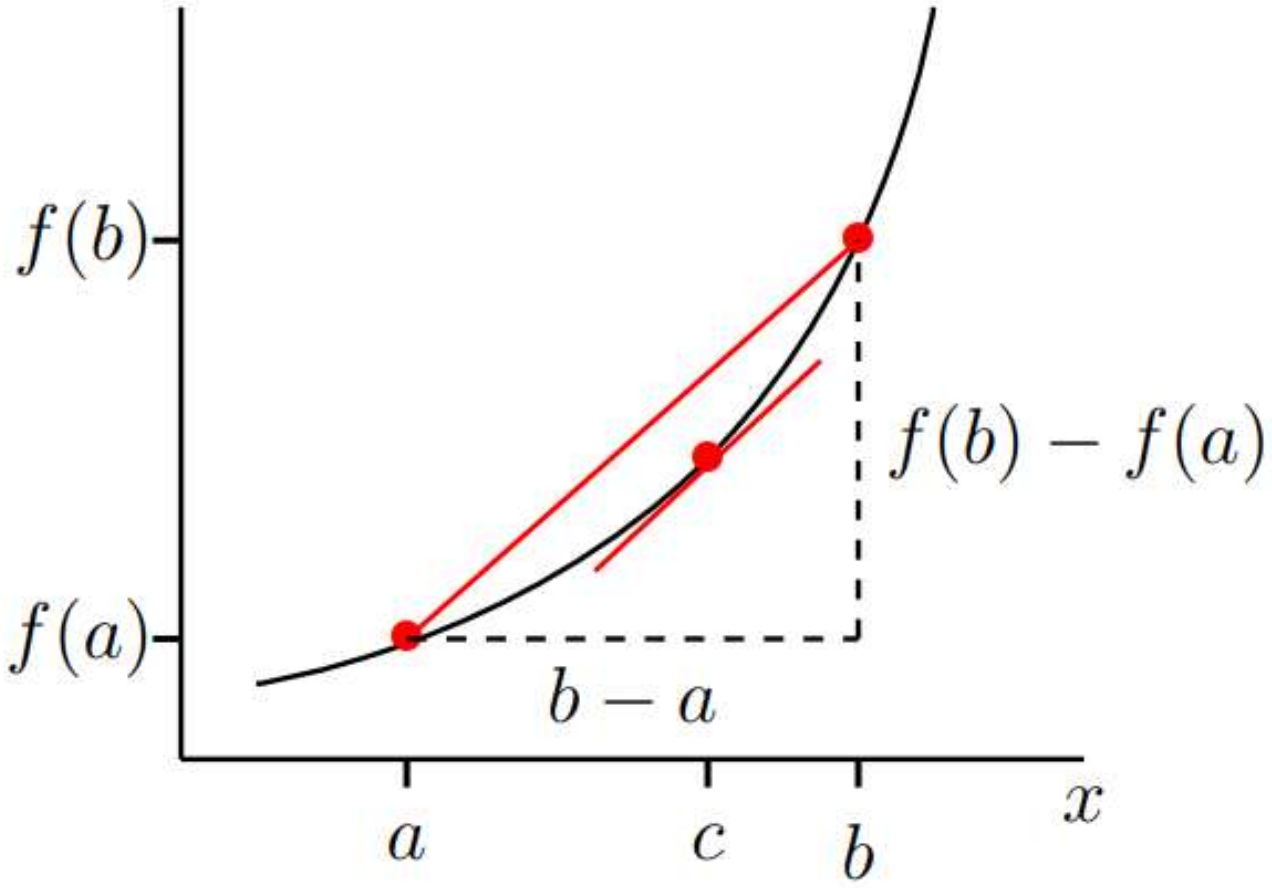
but $f'(c) = 3c^2 + 1 > 0$, a contradiction.

Mean Value Theorem

Theorem 4.4. Mean Value Theorem

Let $f : \left[a, b\right] \to \mathbb{R}$ be continuous on $\left[a, b\right]$ and differentiable on $\left(a, b\right)$.

Then there exists a point $c \in \left(a, b\right)$ such that $f'(c) = \frac{f(b)-f(a)}{b-a}$.


$f(b)$
$f(a)$
$f(b) - f(a)$
$b - a$
a
c
b
x

Proof. The equation of the straight line going through the points $\Big(a, f(a)\Big)$ and $\Big(b, f(b)\Big)$ is given by

$$y = \tfrac{f(b)-f(a)}{b-a}(x-a) + f(a)\,. \qquad (1)$$

Now, the (vertical) distance function $d : \Big[a, b\Big] \to \mathbb{R}$ between $f(x)$ and the line y (equation (1)) is given by

$$d(x) = f(x) - \Big[\tfrac{f(b)-f(a)}{b-a}(x-a) + f(a)\Big].$$

Clearly the distance between the curve and the line is zero at the end points, i.e., $d(a) = d(b) = 0$.

Additionally, by Theorem 3.3, and the algebraic properties of differentiable functions, it follows that d is differentiable on $\left(a, b\right)$ and d is continuous on $\left[a, b\right]$, i.e.,

$$d'(x) = f'(x) - \tfrac{f(b)-f(a)}{b-a}\,. \qquad (2)$$

$$d'(x) = f'(x) - \frac{f(b)-f(a)}{b-a}\,. \qquad (2)$$

By Theorem 4.2 (RT), $\exists c \in \Big(a,b\Big)$ such that

$d'(c) = 0$ in (2), i.e., $f'(c) = \frac{f(b)-f(a)}{b-a}\,.$

True or False

RT is a special case of MVT.

True. $f(a) = f(b) \Rightarrow f'(c) = 0\,.$

Increasing/decreasing functions

Definition (increasing/decreasing function)

A function $f : D \to \mathbb{R}$ is increasing on D if

$$f(x) \leq f(y) \text{ whenever } x < y \text{ in } D,$$

and decreasing if

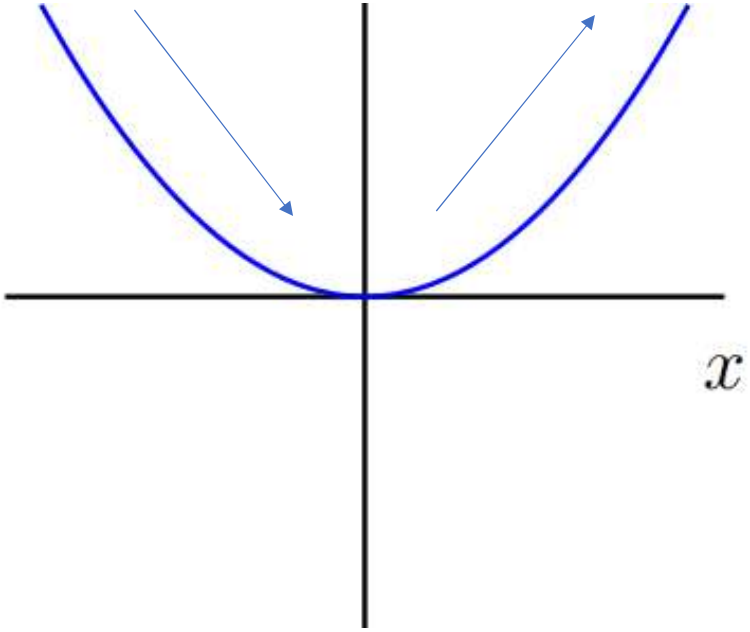
$$f(x) \geq f(y) \text{ whenever } x < y \text{ in } D.$$

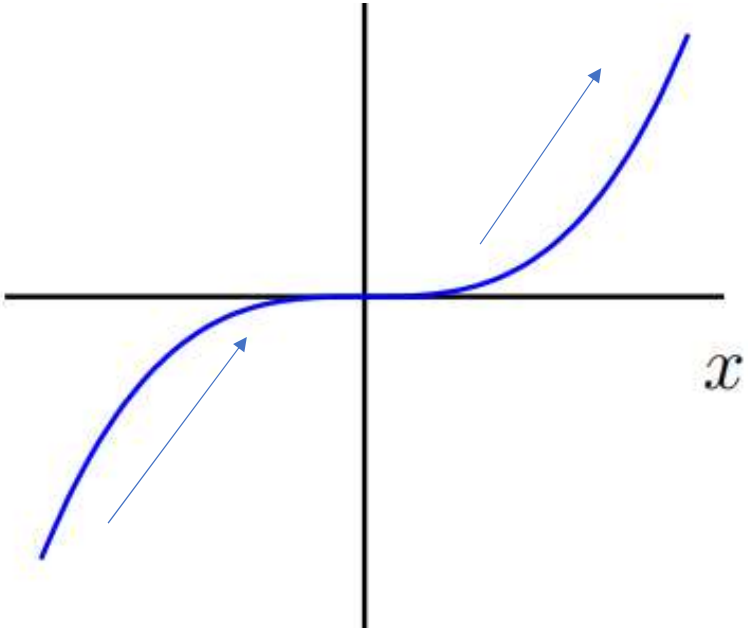
Remarks: 1. A monotone function is one that is either increasing or decreasing.

2. f is strictly increasing on D if $x < y \Rightarrow f(x) < f(y)$.

Example

The function $f(x) = x^3$ is increasing on $\mathbb{R}$. The function $g(x) = x^2$ is neither increasing nor decreasing on $\mathbb{R}$; it is decreasing on $(-\infty, 0]$ and increasing on $[0, \infty)$.





Theorem. Increasing/decreasing function

Let $f : D \to \mathbb{R}$ be differentiable on D.

Then f is increasing on D if and only if

$$f' \geq 0 \text{ on } D.$$

$\Rightarrow$) Suppose f is increasing. Since $c \in \left(a, b\right)$,

we construct two sequences $\left(x_n\right)$ and $\left(y_n\right)$,

which converge to c and satisfy $x_n < c < y_n \forall n \in \mathbb{N}$.

Now $\quad x_n < c \Rightarrow f(x_n) \leq f(c)\ \forall n \in \mathbb{N}$ and

$\qquad\qquad c < y_n \Rightarrow f(c) \leq f(y_n)\ \forall n \in \mathbb{N}$.

Thus $$f'(c) = \lim_{n\to\infty} \frac{f(y_n)-f(c)}{y_n-c} \geq 0 \text{ and}$$

$$f'(c) = \lim_{n\to\infty} \frac{f(x_n)-f(c)}{x_n-c} \geq 0$$

by the Order Limit Theorem (Theorem 6.2, Chap 2).

$\Leftarrow$) Suppose $f'(x) \geq 0 \ \forall x \in D$. Let $a, b \in D$ with $a < b$.

By the MVT, $\exists c \in \Big(a, b\Big) \ni f(b) - f(a) = f'(c)(b - a)$.

Since $f'(c) \geq 0$ and $b > a$, it follows that $f(b) \geq f(a)$.

Remark. f is decreasing on D iff $f' \leq 0$ on D.

We leave the proof as an exercise for the reader.

Remark

In Q14, WS6, you will prove that

$f'(x) \neq 0 \ \forall x \in I \Rightarrow f$ is injective on I,

but the converse is not true.

This leads to the following outcomes:

- f is increasing on $I \Leftrightarrow f' \geq 0$ on I.
- f is strictly increasing on $I \Rightarrow f' \geq 0$ on I.
- f is strictly increasing on $I \Leftarrow f' > 0$ on I.

Example

$f(x) = x^3$ is increasing on $\mathbb{R}$

because $f'(x) = 3x^2 \geq 0$.

$g(x) = x^2$ is decreasing on $\mathbb{R} \setminus \mathbb{R}^+$

because $g'(x) = 2x \leq 0$.

Example

Prove that $\left(\sin\left(\frac{1}{k}\right)\right),\ k \in \mathbb{N}$ is decreasing.

Proof. Note that $f(x) = \sin x$ is increasing on $\left[0, \frac{\pi}{2}\right]$ because $f'(x) = \cos x \geq 0\ \forall x \in \left[0, \frac{\pi}{2}\right]$.

$\forall k \in \mathbb{N},\ 0 < \frac{1}{k} \leq 1 < \frac{\pi}{2} \Rightarrow \sin\left(\frac{1}{k+1}\right) \leq \sin\left(\frac{1}{k}\right)$

and so $\left(\sin\left(\frac{1}{k}\right)\right)$ is decreasing.

True or False

$f' = 0$ if and only f is a constant function.

In general, f is a constant function $\Rightarrow$ $f' = 0$ is true.

False (at a point):

$f(x) = x^3$ is not a constant function but $f'(0) = 0$.

True (on an interval):

$f' = 0$ on an interval iff f is constant on that interval.

Applications of increasing/decreasing function

Example

Prove that

(a) $\ln(1+x) \geq \frac{1}{2}x \ \forall x \in \left[0,1\right].$

(b) $e^x \geq 1+x \ \forall x \in \mathbb{R}\,.$

Proof

(a) Let $f(x) = \ln(1+x) - \frac{1}{2}x \ \forall x \in \left[0,1\right]$.

Then f is continuous on $\left[0,1\right]$ and differentiable on $\left(0,1\right)$, and $f'(x) = \frac{1-x}{2(1+x)} \geq 0 \ \forall x \in \left(0,1\right)$.

Thus, f is increasing on $\left(0,1\right)$. So,

$$0 < x \Rightarrow f(0) \leq f(x) \Rightarrow \ln(1+x) \geq \frac{1}{2}x\,.$$

Proof. (b) Let $g(x) = e^x - 1 - x$.

Then g is continuous on $\mathbb{R}$, differentiable on $\mathbb{R}$,

and $g'(x) = e^x - 1 \geq 0$ for all $x \geq 0$,

but $g'(x) \leq 0$ for all $x \leq 0$.

Thus, g is increasing on $\mathbb{R}^+$ and decreasing on $\mathbb{R}^-$.

Note that e^x is increasing for all $\mathbb{R}$.

Case 1. Let $x > 0$. Then $g(x) \geq g(0) \Rightarrow e^x - 1 - x \geq 0$.

Case 2. Let $x = 0$. Then the equality holds.

Case 3. Let $x < 0$. Then $g(x) \geq g(0) \Rightarrow e^x - 1 - x \geq 0$.

Applications of MVT

Example 4.5(i): Apply MVT on inequality

Prove that $\left|\cos x - 1\right| \leq \left|x\right|$ for all $x \in \mathbb{R}$.

Proof. The equality holds true when $x = 0$.

We will consider two cases: $x > 0$ and $x < 0$.

Case 1. $x > 0$. Let $f : [0, \infty) \to \mathbb{R}$ be function defined by $f(x) = \cos x$. Then f is continuous on $[0, x]$ and differentiable on $(0, x)$, it follows by MVT that $\exists c \in (0, x)$ such that $-\sin c = f'(c) = \frac{f(x)-f(0)}{x-0} = \frac{\cos x - 1}{x}$. Since $|\sin c| \leq 1$, it follows that $|\cos x - 1| \leq |x|$.

Case 2. $x < 0$. Let $g : \left(-\infty, 0\right] \to \mathbb{R}$ be function defined by $g(x) = \cos x$. Then g is continuous on $\left[x, 0\right]$ and differentiable on $\left(x, 0\right)$, it follows by MVT that $\exists c \in \left(x, 0\right)$ such that $-\sin c = g'(c) = \frac{g(0)-g(x)}{0-x} = \frac{1-\cos x}{-x} = \frac{\cos x - 1}{x}$.

Since $\left|\sin c\right| \leq 1$, it follows that $\left|\cos x - 1\right| \leq \left|x\right|$.

Extra example

Recall in Example 4.4, Chapter 4:

We proved $\left|\sin x\right| \leq \left|x\right|$ for all $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

Now we want to prove that the result is true for all $x \in \mathbb{R}$.

Proof. The equality holds when $x = 0$.

Case 1. When $x > 0$. Let $f : [0, \infty) \to \mathbb{R}$ be function given by $f(x) = \sin x$. Note that f is continuous on $\left[0, x\right]$ and differentiable on $\left(0, x\right)$. Then by MVT, $\exists c \in \left(0, x\right)$ such that $\cos c = f'(c) = \frac{f(x) - f(0)}{x} = \frac{\sin x}{x}$.

Now $\left|\cos c\right| \leq 1$ implies that $\left|\sin x\right| \leq \left|x\right|$.

Case 2. When $x < 0$. Let $g : (-\infty, 0] \to \mathbb{R}$ be function given by $f(x) = \sin x$. Note that f is continuous on $\big[x, 0\big]$ and differentiable on $\big(x, 0\big)$. Then by MVT, $\exists\, c \in \big(x, 0\big)$ such that $\cos c = f'(c) = \frac{\sin x}{x}$.

Now $\big|\cos c\big| \leq 1$ implies that $\big|\sin x\big| \leq \big|x\big|$.

Example 4.5 (ii)

(ii) Let $f : \left(0, \infty\right) \to \mathbb{R}$ be a function given by

$$f(x) = x \ln x \, .$$

Prove that for each $k \in \mathbb{N}$,

$$(k+1)\ln(k+1) - k \ln k \leq 1 + \ln(k+1) \, .$$

Hence, deduce that

$$n \ln n < n + \ln(n!) \; \forall n \in \mathbb{N} \setminus \{1\} \, .$$

Proof. Let $f(x) = x \ln x\,,\ x \geq 1$. Let $k \in \mathbb{N}$. Then f is continuous on $\left[k, k+1\right]$ and differentiable on $\left(k, k+1\right)$. Applying the MVT: $f(k+1) - f(k) = f'(c)$ for some $c \in \left(k, k+1\right)$, i.e., $(k+1)\ln(k+1) - k \ln k = f'(c)\,.$

Note that $f'(x) = 1 + \ln x$ is increasing on $\Big(k, k+1\Big)$ because $f''(x) = \frac{1}{x} > 0$ on $\Big(k, k+1\Big)$.

Then $k < c < k+1 \Rightarrow f'(k) \leq f'(c) \leq f'(k+1)$. So,

$$1 + \ln k \leq (k+1)\ln(k+1) - k \ln k \leq 1 + \ln(k+1).$$

Now, for $n \in \mathbb{N} \setminus \{1\}$, take the sum of these inequalities from $k = 1$ to $k = n - 1$; and the proof is completed:

$$n \ln n = f(n) - f(1) = \sum_{k=1}^{n-1} \Big[f(k+1) - f(k) \Big]$$

$$\leq \sum_{k=1}^{n-1} \Big[\ln(r+1) + 1 \Big] = \ln(n!) + n - 1 < \ln(n!) + n$$

Remark: Note that $n \neq 1$ or otherwise, from $k = 1$ to $k = 0$, which is absurd.

Cauchy Mean value theorem

Theorem 4.6. Cauchy's Mean Value Theorem

If functions f and g are continuous on $\left[a,b\right]$ and differentiable on $\left(a,b\right)$, then $\exists c \in \left(a,b\right)$ such that $\Big(f(b)-f(a)\Big)g'(c) = \Big(g(b)-g(a)\Big)f'(c)$.

Remark: If $g(b)-g(a) \neq 0 \neq g'(c)$, then $\frac{f'(c)}{g'(c)} = \frac{f(b)-f(a)}{g(b)-g(a)}$.

Proof. The case $g(a) = g(b)$ is immediate since this implies that $g'(c) = 0$ by RT and hence the equality $\Big(f(b) - f(a)\Big)g'(c) = \Big(g(b) - g(a)\Big)f'(c)$ holds true.

Assume that $g(a) \neq g(b)$. Define

$$h(x) = f(x) - \frac{f(b)-f(a)}{g(b)-g(a)} \cdot g(x).$$

Clearly $h(a) = h(b)$. Apply RT:

$\exists c \in \Big(a, b\Big)$ such that

$$0 = h'(c) = f'(c) - \frac{f(b)-f(a)}{g(b)-g(a)} \cdot g'(c)$$

Hence, $\Big(f(b) - f(a)\Big)g'(c) = \Big(g(b) - g(a)\Big)f'(c)$.

Verify that $h(x) = f(x) - \frac{f(b)-f(a)}{g(b)-g(a)} \cdot g(x) \Rightarrow h(a) = h(b)$.

$$h(b) = f(b) - \frac{f(b)-f(a)}{g(b)-g(a)} \cdot g(b)$$

$$(-) \qquad h(a) = f(a) - \frac{f(b)-f(a)}{g(b)-g(a)} \cdot g(a)$$

$$h(b) - h(a) = f(b) - f(a) - \frac{f(b)-f(a)}{g(b)-g(a)}\Big(g(b) - g(a)\Big) = 0$$

Remark: The classical MVT is a special case of CMVT (when $g(x) \equiv x$). So, it is also called the Generalised MVT.

Applications of CMVT

L'Hôpital's rule.

Let $f, g : \left[a, b\right] \to \mathbb{R}$ be continuous functions on $\left[a, b\right]$ and differentiable on $\left(a, b\right)$.

Let $\alpha \in \left[a, b\right]$ such that $f(\alpha) = g(\alpha) = 0$.

Then $\lim\limits_{x \to \alpha} \frac{f'(x)}{g'(x)} = L \Rightarrow \lim\limits_{x \to \alpha} \frac{f(x)}{g(x)} = L$.

Proof. Given $\lim\limits_{x\to\alpha}\frac{f'(x)}{g'(x)}=L$, i.e., $\lim\limits_{x\to\alpha^-}\frac{f'(x)}{g'(x)}=L=\lim\limits_{x\to\alpha^+}\frac{f'(x)}{g'(x)}$.

Consider $\lim\limits_{x\to\alpha^+}\frac{f'(x)}{g'(x)}=L$, then $\forall\varepsilon>0\ \exists\delta>0$ such that

$$\forall x\in\Big(\alpha,\alpha+\delta\Big)\Rightarrow\left|\frac{f'(x)}{g'(x)}-L\right|<\varepsilon\,.$$

Apply CMVT on $\Big(\alpha, x\Big)$ to obtain

$$\frac{f'(c)}{g'(c)} = \frac{f(x)-f(\alpha)}{g(x)-g(\alpha)}$$

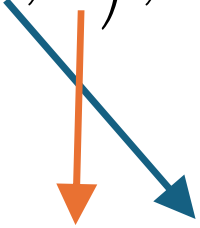
for some $c \in \Big(\alpha, x\Big) \subset \Big(\alpha, \alpha+\delta\Big)$. Hence,

$$\left|\frac{f(x)}{g(x)} - L\right| = \left|\frac{f(x)-f(\alpha)}{g(x)-g(\alpha)} - L\right| = \left|\frac{f'(c)}{g'(c)} - L\right| < \varepsilon\,.$$

This proves that $\lim\limits_{x\to\alpha^+} \frac{f(x)}{g(x)} = L\,.$

Similarly, when $\lim_{x \to a^-} \frac{f'(x)}{g'(x)} = L$, we can prove that $\lim_{x \to a^-} \frac{f(x)}{g(x)} = L$, and thus implying $\lim_{x \to a} \frac{f(x)}{g(x)} = L$, as desired.

Remark: We choose to apply CMVT on $\left(\alpha, x\right)$ because then $c \in \left(\alpha, x\right)$, such that



$$\frac{f'(c)}{g'(c)} = \frac{f(x)-f(\alpha)}{g(x)-g(\alpha)}.$$

If we apply CMVT on $\left(\alpha, \alpha+\delta\right)$, then we will get

$$\frac{f'(c)}{g'(c)} = \frac{f(\alpha+\delta)-f(\alpha)}{g(\alpha+\delta)-g(\alpha)}.$$

Example

Use *L'Hôpital's rule* to find the following limits.

(a) $\lim\limits_{x\to 0} \frac{\ln(x+1)}{x}$ (b) $\lim\limits_{x\to 0}(\tan x \csc x)$

(c) $\lim\limits_{x\to 0^+} \frac{\ln(\ln(x+1)+1)}{x}$ (d) $\lim\limits_{x\to 0} \frac{1-\cos x}{x^2}$

Solution. (a) Since $\ln 1 = 0$, it follows that

$$\lim_{x\to 0} \frac{\ln(x+1)}{x} = \lim_{x\to 0} \frac{1}{x+1} = 1.$$

(b) Since $\tan 0 = 0 = \sin 0$, it follows that

$$\lim_{x\to 0} \frac{\tan x}{\sin x} = \lim_{x\to 0} \frac{\sec^2 x}{\cos x} = \lim_{x\to 0} \frac{1}{\cos^3 x} = 1.$$

(c) Since $\ln(\ln 1 + 1) = 0$, it follows that

$$\lim_{x\to 0^+} \frac{\ln(\ln(x+1)+1)}{x} = \lim_{x\to 0^+} \frac{1/(x+1)}{\ln(x+1)+1} = 1$$

Remark: L'Hôpital's rule can be applied to one sided limit.

(d) Since $1-\cos 0 = 0 = 0^2$, it follows that

$$\lim_{x\to 0} \frac{1-\cos x}{x^2} = \lim_{x\to 0} \frac{\sin x}{2x} = \frac{1}{2}\lim_{x\to 0} \frac{\sin x}{x}.$$

Either we can conclude that the above limit is ½ because we already know $\lim_{x\to 0} \frac{\sin x}{x} = 1$ or we may apply L'Hôpital's rule multiple times to obtain the final answer.

Another application of CMVT

Lagrange Remainder Theorem
(the most demanding proof of F18CD).